

Unit 10, Lesson 7: Measures of Center and Spread

Benchmark

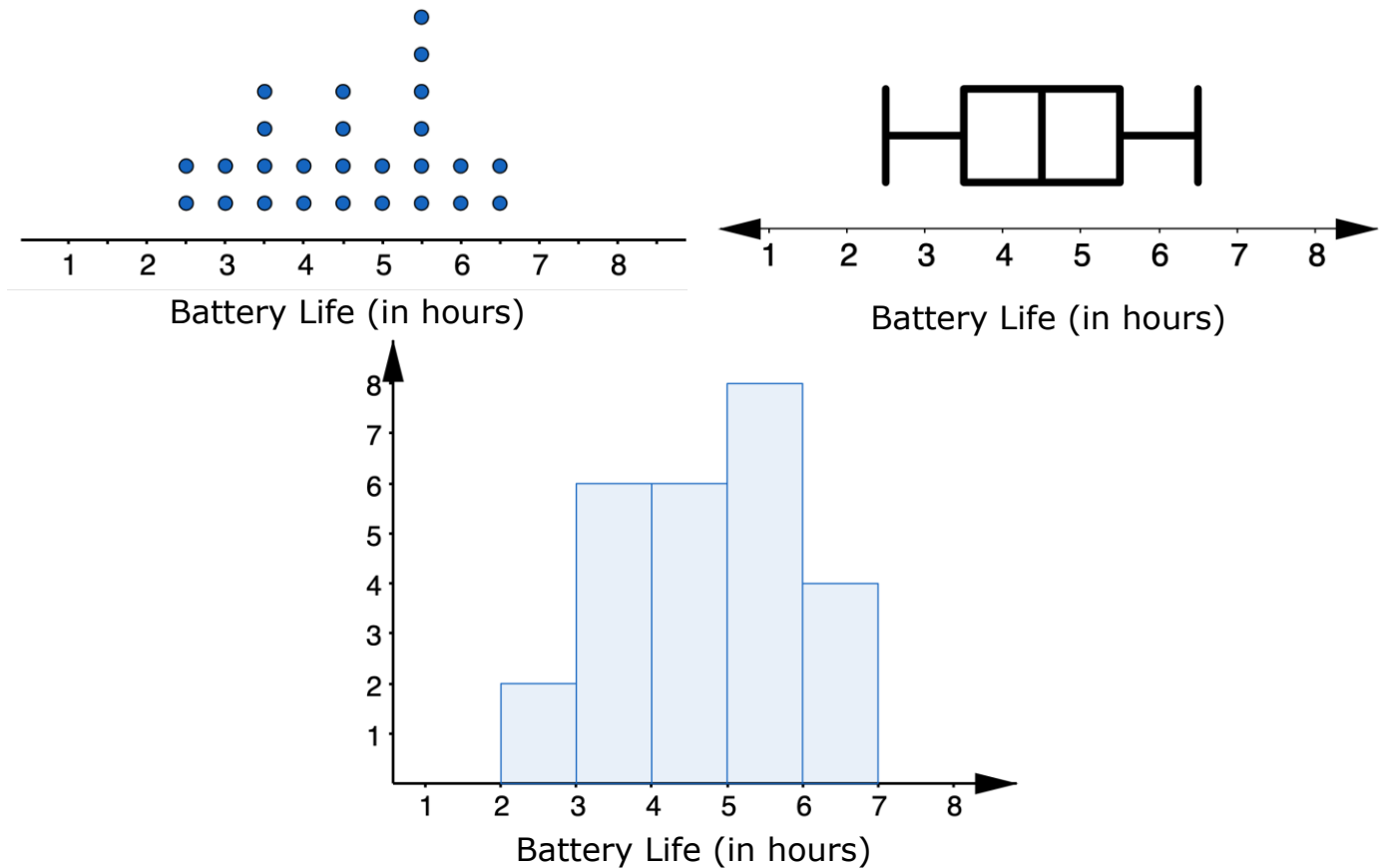
MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

Learning Target

- ☐ I can determine which measures of center and variation best describe a distribution.

10.7.1 Warm-Up: Notice and Wonder

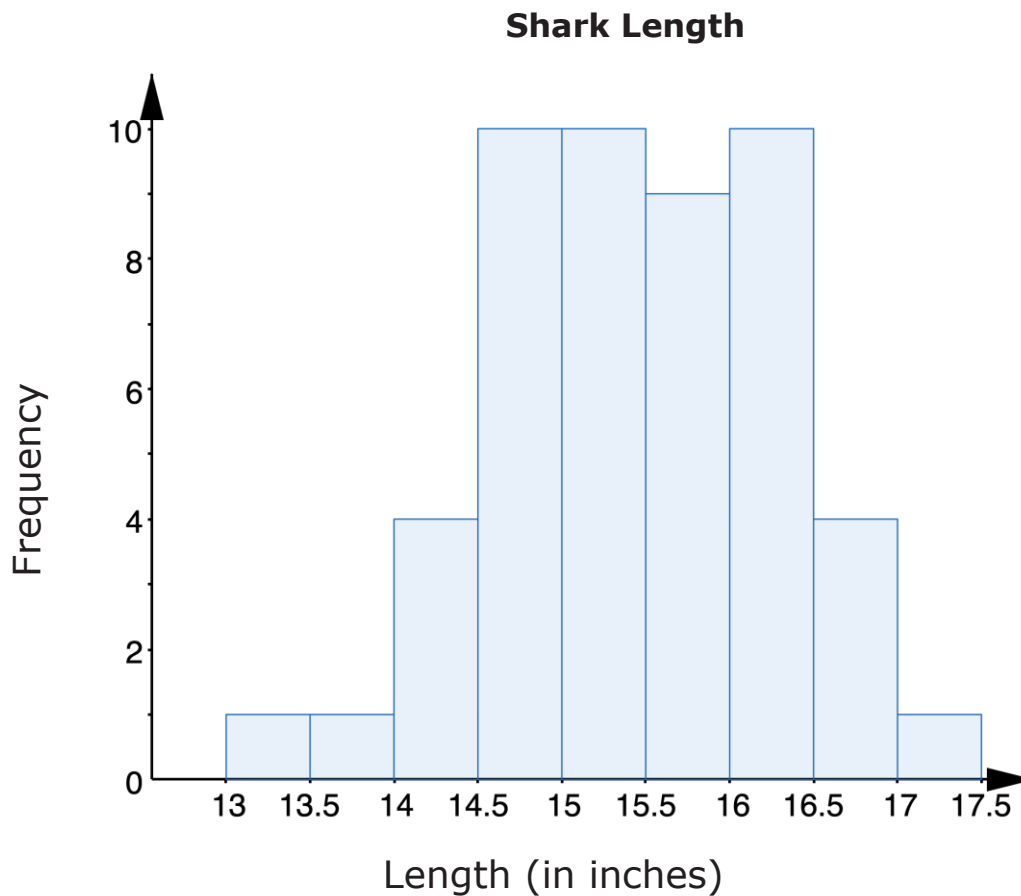
The dot plot, histogram, and box plot summarize the hours of battery life for 26 cell phones that are constantly streaming videos.



1. What do you notice?
2. What do you wonder?

10.7.2 Exploration: Measures of Center

1. Consider the histogram of the lengths of female sharks.



- Describe the shape of the histogram.
- Circle the value on the horizontal axis of the graph that shows the approximate location of the median of the data.
- Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.

- d. Compare your answers to B and C with your partner's answers. Discuss your reasoning and, if necessary, make changes.



In a symmetrical graph, the mean and the median are located in the middle of the graph, approximately in the same location.

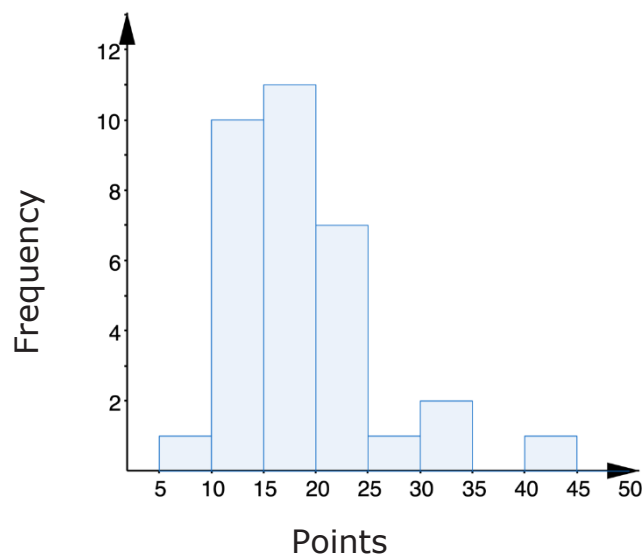
- e. The mean and the median of this histogram are located between 15 – 15.5. Discuss with your partner which measure most accurately describes the center of the histogram. Summarize your discussion.



In a symmetrical graph with no outliers, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution.

2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.

2008 Points per Game



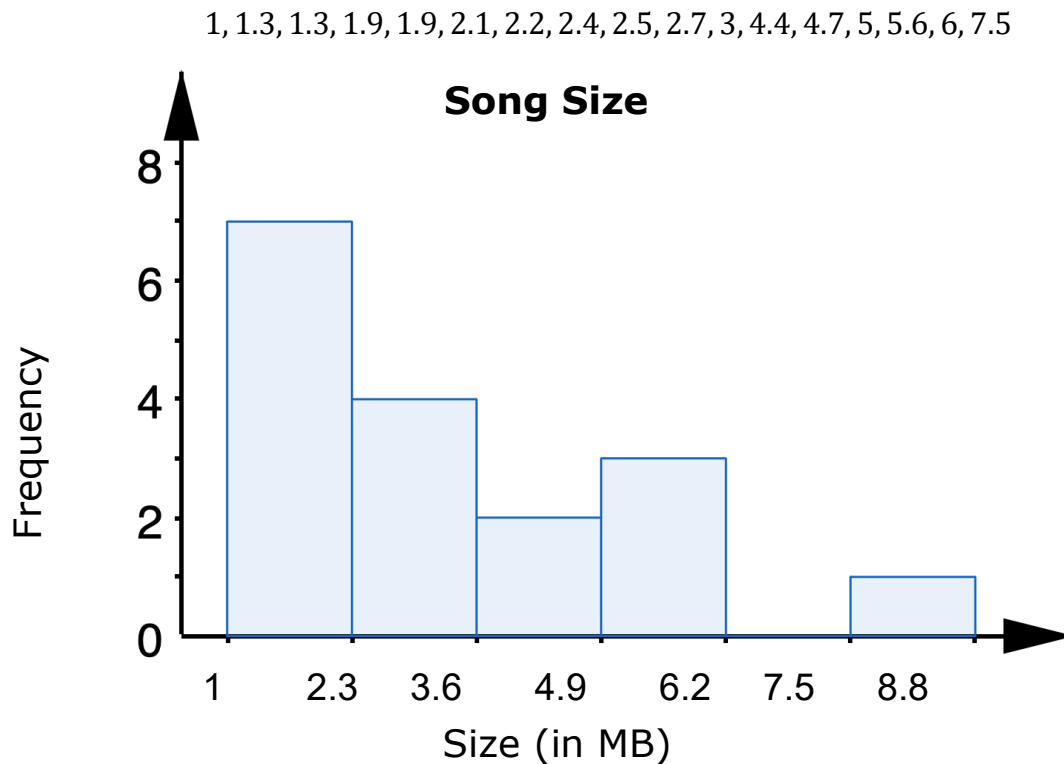
- a. Describe the shape of the histogram.
- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of this data.
- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.
- d. In a symmetrical graph, the mean is the preferred measure of center to use. The histogram for points scored is skewed. Discuss with your partner why the mean would not be a good representation of the center for points scored. Summarize your discussion.
- e. In this graph, the median is located between 15 – 20 and the mean is in the 20 – 25 interval. Which measure most accurately describes the center of the histogram?
- f. Which measure of variation should be used to describe the distribution?



In a skewed graph or a graph with outliers, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution.

10.7.3 Try It: Measures of Center

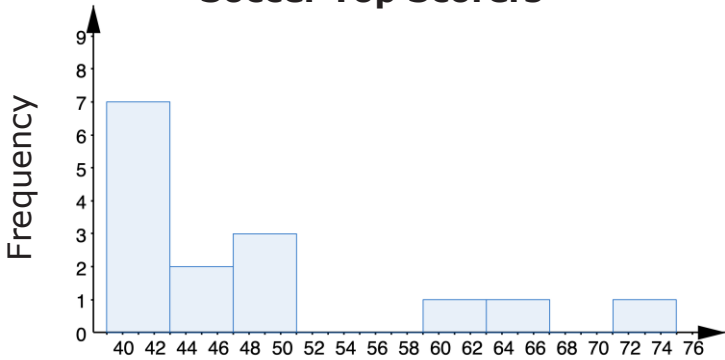
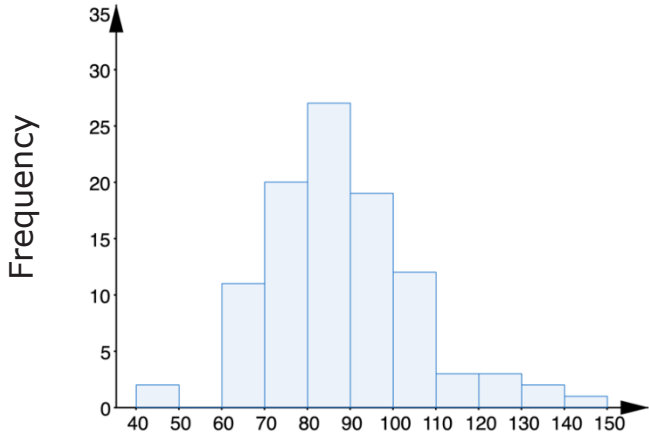
1. Nelson is interested in how much storage space his downloaded music takes up. The sizes of 17 randomly selected files are listed, in megabytes (MB), and displayed in a histogram.



- Describe the shape of the histogram.
- Which measure of center best describes the center of the histogram?
- Explain your choice of center.

10.7.4 Your Turn!

1. Determine which measures of center and variation best describe each data set.

Data	Measure of Center	Measure of Variation
Weekly biking times, in hours: 1,2,2,3,3,3,4,4,4,4,5,5,6,6,7	☐ Mean ☐ Median	☐ IQR ☐ Range
Daily high temperatures, in °F: 61, 61, 62, 64, 66, 67, 67, 67, 68, 69, 70, 70, 70, 71, 71, 72, 74, 74, 74, 75, 75, 75, 76, 76, 77, 78, 78, 79, 79, 95	☐ Mean ☐ Median	☐ IQR ☐ Range
2019-2020 NCAA Women's Division 1 Soccer Top Scorers  Frequency Number of Goals	☐ Mean ☐ Median	☐ IQR ☐ Range
Text Messages Sent  Frequency Number of Text Messages	☐ Mean ☐ Median	☐ IQR ☐ Range

10.7.5 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate where your current learning is related to each target.

I can determine which measures of center and variation best describe a distribution.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.

